DEPARTMENT OF COMPUTER SCIENCE
M.Sc. Computer Science (Data Analytics)

ASSIGNMENT – 1

Probability and Bayesian Classification

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1. Probability Concept

1.1 Introduction

Uncertainty is a permanent feature of the natural and social world. Whether one is predicting rainfall, estimating the reliability of a hardware component, or deciding how confident a machine-learning model should be about a prediction, the underlying question is always the same: how likely is a particular outcome to occur? Probability theory is the branch of mathematics that provides a rigorous language for answering this question. It converts vague notions of "likely" and "unlikely" into precise numerical statements that can be manipulated, compared, and used for decision-making.

This chapter builds the conceptual foundation on which the remainder of the assignment rests. Beginning with the classical definition of probability, it introduces the vocabulary of experiments, outcomes, and events, before moving on to the classification of events that recurs throughout every subsequent topic — the addition rule, the multiplication rule, conditional probability, the law of total probability, and ultimately Bayes' Theorem and its application in Bayesian classification.

1.2 Definition

Probability is a numerical measure, lying between 0 and 1 inclusive, that quantifies the likelihood of occurrence of an event associated with a random experiment. The classical (or a priori) definition, attributed to Pierre-Simon Laplace, states that if a random experiment has n(S) equally likely, mutually exclusive outcomes, and an event A is favoured by n(A) of these outcomes, then the probability of A is defined as follows.

$$P(A) = n(A) / n(S) = \text{Number of favourable outcomes} / \text{Total number of possible outcomes}$$

A probability of 0 indicates that the event is impossible, while a probability of 1 indicates that the event is certain. Every other real number between these two extremes reflects a graded degree of belief or frequency of occurrence.

1.3 Importance of Probability

Probability theory is not merely an abstract mathematical exercise; it is the quantitative backbone of decision-making under uncertainty. Its importance can be appreciated through the following observations.

- It allows engineers to design systems — such as communication networks and manufacturing lines — that behave reliably despite random fluctuations and component failures.
- It underlies statistical inference, enabling researchers to draw conclusions about an entire population from a limited sample.
- It forms the mathematical foundation of machine learning algorithms, including the Naïve Bayes classifier discussed later in this assignment, which explicitly reasons in terms of probabilities.
- It supports risk assessment in finance, insurance, and medicine, where decisions must be made despite incomplete information.
- It provides the theoretical basis for randomised algorithms and simulation techniques widely used in computer science.

1.4 Basic Terminology

Before any formula can be meaningfully applied, it is essential to be precise about the vocabulary of probability theory. The following terms recur throughout this assignment and must be clearly distinguished from one another.

1.4.1 Experiment

An experiment is any well-defined procedure or process that produces an observable result. Measuring the height of a student, tossing a coin, or recording the number of defective items in a batch are all examples of experiments.

1.4.2 Random Experiment

A random experiment is an experiment whose outcome cannot be predicted with certainty in advance, even though all of its possible outcomes are known beforehand, and which can, in principle, be repeated under identical conditions. Tossing a fair coin is a random experiment because, although we know the outcome will be either a head or a tail, we cannot say with certainty which one will occur on a given trial.

1.4.3 Outcome

An outcome is a single possible result of a random experiment. When a die is rolled once, obtaining a "4" is one particular outcome among the six that are possible.

1.4.4 Event

An event is a subset of the sample space, that is, a collection of one or more outcomes that share a property of interest. For instance, when a die is rolled, the event "an even number appears" corresponds to the subset $\{2, 4, 6\}$.

1.4.5 Sample Space

The sample space, usually denoted S, is the set of every possible outcome of a random experiment. For a single coin toss, S = {H, T}; for a single die roll, S = {1, 2, 3, 4, 5, 6}.

1.4.6 Equally Likely Events

Two or more outcomes are said to be equally likely if there is no reason, physical or otherwise, to expect one to occur more often than another. Each face of an unbiased die is equally likely to appear on any given roll.

1.4.7 Impossible Event

An impossible event is one that can never occur under the conditions of the experiment; it corresponds to the empty set, and its probability is always zero. Obtaining a "7" on a single roll of a standard six-faced die is an impossible event.

1.4.8 Certain Event

A certain (or sure) event is one that is bound to occur every time the experiment is performed; it corresponds to the entire sample space, and its probability is always one. Obtaining a number less than 7 on a single die roll is a certain event.

1.4.9 Independent Events

Two events are independent if the occurrence of one has absolutely no bearing on the probability of occurrence of the other. The outcome of one coin toss, for example, does not influence the outcome of a subsequent toss.

1.4.10 Dependent Events

Two events are dependent if the occurrence of one event alters the probability of the other. Drawing two cards in succession from a deck without replacement produces dependent events, because the composition of the deck changes after the first draw.

1.4.11 Mutually Exclusive Events

Two events are mutually exclusive (or disjoint) if they cannot occur simultaneously in a single trial of the experiment, that is, their intersection is the empty set. On a single die roll, the events "an even number appears" and "an odd number appears" are mutually exclusive.

1.5 Formula

$$P(A) = n(A) / n(S), \qquad 0 \le P(A) \le 1$$

1.6 Explanation of Formula

The formula expresses probability as a ratio of two counts. The numerator, n(A), counts only those outcomes in the sample space that satisfy the condition defining event A; the denominator, n(S), counts every outcome that could possibly occur, regardless of whether it satisfies A. Because n(A) can never exceed n(S), and n(A) can never be negative, the resulting ratio is always confined to the closed interval [0, 1]. A ratio close to 1 indicates that the event occupies a large proportion of the sample space and is therefore highly likely; a ratio close to 0 indicates the opposite.

1.7 Real-life Applications

- Weather forecasting agencies express the chance of rainfall as a probability derived from historical and current atmospheric data.
- Insurance companies use probability to price policies based on the likelihood of claims such as accidents or illness.
- Quality-control engineers use probability to estimate the proportion of defective items in a production batch.
- Search engines and recommendation systems use probabilistic models to rank the relevance of results for a given user.
- Medical diagnostic tests report probabilities of disease presence given a positive or negative test result.

1.8 Summary

This section established the classical definition of probability and the precise vocabulary — experiment, outcome, event, and sample space — required to apply it correctly. It also distinguished between several important classifications of events, namely equally likely, impossible, certain, independent, dependent, and mutually exclusive events. These distinctions are not merely academic; the correct choice of formula in every subsequent section of this assignment depends on correctly identifying which of these categories a given pair of events belongs to.

1.9 Diagrams

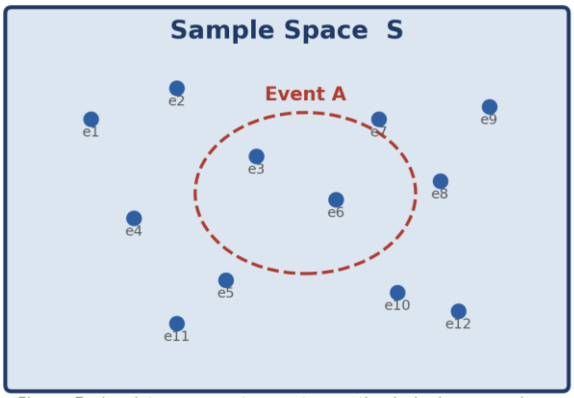

Figure: Each point e_i represents an outcome; the dashed curve encloses the outcomes that form Event A within the Sample Space S.

Figure 1: *A generic sample space S containing individual outcomes e_1, e_2, …, e_{12}, with Event A shown as the subset of outcomes enclosed by the dashed curve.*

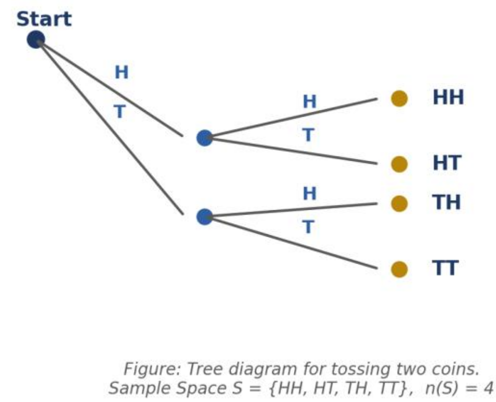

Figure 2: *Sample space for tossing two fair coins in succession, obtained via a tree diagram. S = {HH, HT, TH, TT}.*

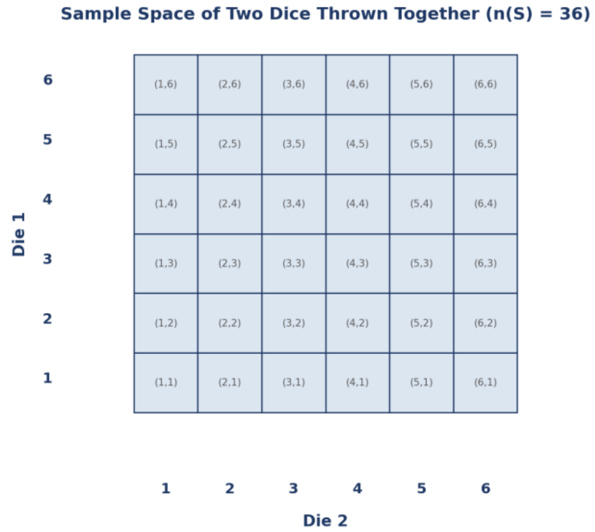

Figure 3: *Sample space for throwing two dice together, shown as a 6 × 6 grid of ordered pairs (Die 1, Die 2). n(S) = 36.*

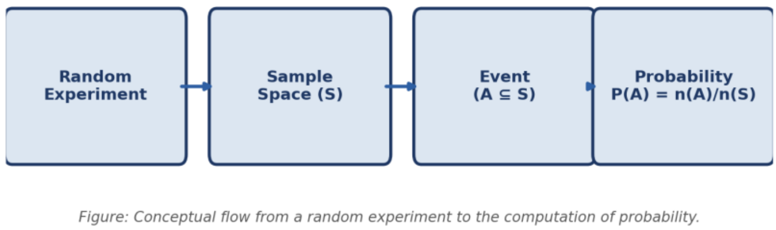

Figure 4: *Conceptual flow diagram showing how probability is computed, starting from a random experiment and ending with the numerical value P(A).*

1.10 Solved Example

(Original University-Level Question)

Question: A box contains 8 red balls, 5 green balls, and 7 blue balls, all identical in size and texture. A ball is drawn at random from the box. Find the probability that the ball drawn is (a) red, and (b) not blue.

Given Data

Number of red balls	8
Number of green balls	5
Number of blue balls	7
Total number of balls	8 + 5 + 7 = 20

What is to be Found

(a) The probability that the ball drawn is red, denoted P(Red). (b) The probability that the ball drawn is not blue, denoted P(Not Blue).

Formula Used

Since every ball is equally likely to be drawn, the classical definition of probability applies directly.

$$P(A) = n(A) / n(S)$$

Solution

Step 1 — Determine the total number of outcomes. Since the box contains 8 + 5 + 7 = 20 balls in total, and each ball is equally likely to be picked, n(S) = 20.

Step 2 — Compute P(Red). The number of favourable outcomes for drawing a red ball is n(Red) = 8, since there are 8 red balls in the box. Substituting into the formula:

$$P(\text{Red}) = n(\text{Red}) / n(S) = 8 / 20 = 2/5 = 0.40$$

Step 3 — Compute P(Not Blue). Rather than counting red and green balls separately, it is more efficient to use the complement rule, since "not blue" is the complement of "blue". The number of blue balls is 7, so P(Blue) = 7/20. Applying the complement rule P(Not Blue) = 1 − P(Blue):

$$P(\text{Not Blue}) = 1 - (7/20) = 13/20 = 0.65$$

This result can be verified directly: the number of balls that are not blue is 8 (red) + 5 (green) = 13, so n(Not Blue)/n(S) = 13/20, which confirms the answer obtained using the complement rule.

Final Answer: P(Red) = 2/5 = 0.40 and P(Not Blue) = 13/20 = 0.65

Exam Tip: *When a question asks for the probability of "not" an event, it is almost always faster to compute the probability of the event itself and subtract from 1, rather than counting every favourable outcome for the complement directly.*

2. Addition Rule of Probability

2.1 Introduction

Many practical questions ask for the probability that at least one of two (or more) events occurs — for instance, the probability that a randomly chosen student has passed Mathematics or Statistics. The Addition Rule (also called the Additive Law of Probability) provides the formal mechanism for combining the individual probabilities of such events into the probability of their union, while correctly handling any overlap between them.

2.2 Definition

The Addition Rule states that the probability of the union of two events A and B — that is, the probability that A occurs, or B occurs, or both occur — is equal to the sum of their individual probabilities, minus the probability that both occur simultaneously.

2.3 Formula

$$\boxed{P(A \cup B) \; = \; P(A) + P(B) - P(A \cap B)}$$

For three events A, B, and C, the rule extends to include every pairwise and triple-wise overlap:

$$\boxed{P(A \cup B \cup C) = P(A)+P(B)+P(C) - P(A\cap B) - P(B\cap C) - P(A\cap C) + P(A\cap B\cap C)}$$

2.4 Explanation

When the probabilities P(A) and P(B) are simply added together, every outcome that belongs to both A and B is counted twice — once as part of A and once again as part of B. The term $P(A \cap B)$ is therefore subtracted once to correct for this double counting, so that every outcome in the union is counted exactly once. This logic mirrors the Inclusion–Exclusion Principle used in set theory and combinatorics.

2.5 When the Addition Rule is Used

2.5.1 Mutually Exclusive Events

If A and B cannot occur together, then $A \cap B$ is the empty set and $P(A \cap B) = 0$. The Addition Rule then simplifies to the special case shown below, which is often the first version of the rule taught to beginners.

$$\boxed{P(A \cup B) = P(A) + P(B) \qquad [\text{when A and B are mutually exclusive}]}$$

2.5.2 Non-Mutually Exclusive Events

If A and B can occur together — that is, if the two events share one or more common outcomes — the general form of the Addition Rule must be used, and the overlap $P(A \cap B)$ must always be subtracted to avoid over-counting. Failing to subtract this term is one of the most common errors made by students new to probability.

2.6 Real-life Applications

- Estimating the probability that a customer purchases either Product A or Product B (or both) in a retail analytics context.
- Computing the probability that a manufactured component fails due to either a mechanical fault or an electrical fault, where both faults may sometimes occur together.
- Determining the probability that a patient tests positive on either of two independent diagnostic screenings.
- Assessing network reliability, where the probability that at least one of several redundant links is available must be computed.

2.7 Diagram

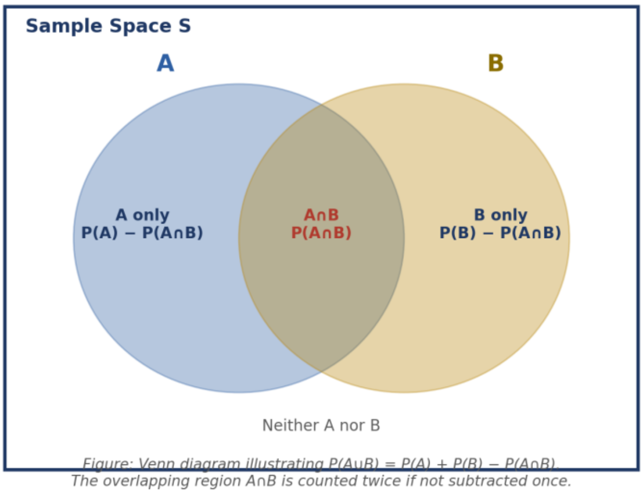

Figure 5: *Venn diagram illustrating the Addition Rule. The overlapping region $A \cap B$ is common to both circles and must be subtracted once from $P(A) + P(B)$ to avoid double counting.*

2.8 Solved Example

(Adapted from GATE Probability Questions, GATE CS 2015)

Question: In a class of 100 students, 45 students have opted for elective Data Structures, 38 students have opted for elective Computer Networks, and 15 students have opted for both electives. A student is chosen at random from the class. Find the probability that the chosen student has opted for at least one of the two electives.

Given Data

Total number of students, n(S)	100
Students who opted Data Structures, n(A)	45

Students who opted Computer Networks, n(B)	38
Students who opted both electives, n(A ∩ B)	15

What is to be Found

The probability that a randomly selected student has opted for at least one of the two electives, that is, $P(A \cup B)$, where A denotes "opted Data Structures" and B denotes "opted Computer Networks".

Formula Used

Since a student may opt for both electives simultaneously, the two events are not mutually exclusive, so the general Addition Rule must be applied.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Solution

Step 1 — Convert each count into a probability by dividing by the total number of students, $n(S) = 100$.

$$P(A) = 45/100 = 0.45, \quad P(B) = 38/100 = 0.38, \quad P(A \cap B) = 15/100 = 0.15$$

Step 2 — Substitute these three probabilities into the Addition Rule formula. The overlap probability $P(A \cap B) = 0.15$ represents the 15 students who opted for both electives and who would otherwise be counted twice.

$$P(A \cup B) = 0.45 + 0.38 - 0.15$$

Step 3 — Carry out the arithmetic. Adding the first two terms gives $0.45 + 0.38 = 0.83$; subtracting the overlap gives the final probability.

$$P(A \cup B) = 0.83 - 0.15 = 0.68$$

Interpretation: 68 out of every 100 students, on average, have opted for at least one of the two electives, while the remaining 32 out of 100 have opted for neither.

Final Answer: $P(A \cup B) = 0.68$ (i.e. 68%)

Exam Tip: *Whenever a GATE-style question mentions students, employees, or items belonging to "either/or/both" categories, immediately suspect a non-mutually-exclusive Addition Rule problem, and look for the phrase "both" to locate the overlap term $P(A \cap B)$.*

3. Multiplication Rule of Probability

3.1 Definition

While the Addition Rule answers questions about the union of events ("A or B"), the Multiplication Rule answers questions about their intersection ("A and B") — the probability that two or more events all occur together in the same trial or across successive trials of an experiment.

3.2 Formula

3.2.1 Independent Events

If two events A and B are independent — meaning the occurrence of one has no effect whatsoever on the probability of the other — their joint probability is simply the product of their individual probabilities.

$$P(A \cap B) = P(A) \times P(B) \qquad \text{[A and B independent]}$$

3.2.2 Dependent Events

If A and B are dependent, the occurrence of A changes the probability that B occurs, and this altered probability, written P(B|A), must be used instead of the unconditional P(B).

$$P(A \cap B) = P(A) \times P(B|A) \qquad \text{[A and B dependent]}$$

3.3 Explanation

The independent-events version of the rule follows from the intuitive idea that if knowing A tells us nothing about B, the chance of both happening together is simply the product of the two separate chances — much as the chance of getting heads on two separate coins is (1/2) × (1/2) = 1/4. The dependent-events version generalises this idea by replacing P(B) with the conditional probability P(B|A), which correctly accounts for the information that A has already occurred and may have changed the sample space relevant to B. This second form is, in fact, the defining relationship from which the formula for conditional probability, discussed in Section 4, is derived algebraically.

3.4 Applications

- Computing the reliability of a series system in which every one of several independent components must function for the system as a whole to work.
- Determining the probability of drawing a specific sequence of cards from a deck without replacement, where each draw changes the composition of the remaining deck.
- Calculating the joint probability of two independent sensor readings both exceeding a threshold in a fault-detection system.

- Modelling the probability that a sequence of independent binary classifier outputs are all correct.

3.5 Diagrams

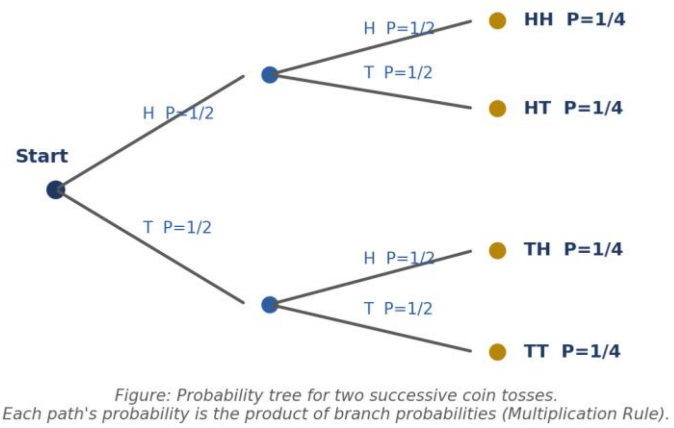

Figure 6: *Probability tree for two successive tosses of a fair coin. Each end-branch probability is obtained by multiplying the probabilities along the path from the root — the Multiplication Rule applied to independent events.*

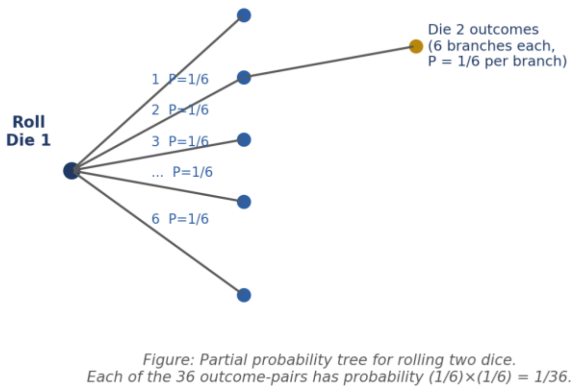

Figure 7: *Partial probability tree for rolling two fair dice in succession, showing that each branch carries probability 1/6, so every ordered outcome pair has joint probability (1/6) × (1/6) = 1/36.*

3.6 Solved Example

(Adapted from GATE Probability Questions, GATE CS 2018)

Question: A bag contains 6 white balls and 4 black balls. Two balls are drawn in succession from the bag without replacement. Find the probability that both balls drawn are black.

Given Data

Number of white balls	6
Number of black balls	4
Total number of balls	10
Sampling method	Without replacement (dependent draws)

What is to be Found

The probability that both balls drawn, in order, are black — that is, P(First ball black ∩ Second ball black).

Formula Used

Because the first ball is not replaced before the second draw, the composition of the bag changes between draws, making the two draws dependent events. The Multiplication Rule for dependent events must therefore be used.

$$P(A \cap B) = P(A) \times P(B|A)$$

Solution

Step 1 — Define events. Let A = "first ball drawn is black" and B = "second ball drawn is black". Since there are 4 black balls out of 10 balls in total at the start:

$$P(A) = 4/10 = 2/5$$

Step 2 — Determine P(B|A). Given that the first ball drawn was black and was not replaced, the bag now contains 9 balls in total, of which only 3 remain black.

$$P(B|A) = 3/9 = 1/3$$

Step 3 — Multiply the two probabilities to obtain the joint probability that both events occur.

$$P(A \cap B) = (2/5) \times (1/3)$$

Step 4 — Carry out the multiplication of the fractions: numerator 2 × 1 = 2, denominator 5 × 3 = 15.

$$P(A \cap B) = 2/15 \; \approx \; 0.1333$$

Final Answer: P(Both balls black) = 2/15 ≈ 0.133

Exam Tip: *The phrase "without replacement" is the standard GATE signal for dependent events. Always recompute both the total count and the favourable count for the second draw before applying P(B|A).*

4. Conditional Probability

4.1 Introduction

In many real situations, probability must be reassessed as new information becomes available. Before a die is rolled, the probability of obtaining a 6 is 1/6; but if we are told, after the roll, that the outcome is even, the probability of it having been a 6 becomes 1/3, because the sample space has effectively shrunk to the three even numbers. Conditional probability is the formal tool for updating a probability estimate in light of such additional information.

4.2 Definition

The conditional probability of an event A, given that another event B has already occurred (with $P(B) > 0$), is denoted $P(A|B)$ and is defined as the probability of the joint occurrence of A and B, divided by the probability of B alone.

4.3 Formula

$$P(A|B) = P(A \cap B) \,/\, P(B)\,, \qquad \text{provided } P(B) > 0$$

By symmetry, the conditional probability of B given A is defined analogously:

$$P(B|A) = P(A \cap B) \,/\, P(A)\,, \qquad \text{provided } P(A) > 0$$

4.4 Explanation

The formula can be understood as a restriction of the sample space. Once it is known that B has occurred, every outcome outside B becomes irrelevant, and B itself effectively becomes the new, reduced sample space. Within this reduced space, the "favourable" outcomes for A are precisely those that lie in both A and B, that is, in $A \cap B$. Dividing $P(A \cap B)$ by $P(B)$ therefore re-normalises the probability so that it is expressed relative to the new, smaller sample space B rather than the original space S.

4.5 Why Conditional Probability is Important

- It is the mathematical foundation of Bayes' Theorem, which in turn underlies the entire family of Bayesian classification algorithms discussed in Section 7.
- It allows probabilistic models to incorporate evidence sequentially, updating beliefs as more data becomes available.
- It is essential in medical diagnostics, where the probability of a disease must be reassessed given a positive or negative test result.
- It underlies the notion of statistical independence itself, since A and B are independent precisely when $P(A|B) = P(A)$.

4.6 Applications

- Spam filters compute the conditional probability that an email is spam, given the presence of specific words.
- Credit-scoring systems compute the conditional probability of default, given a customer's financial history.
- Search engines rank documents by the conditional probability of relevance, given a user's query terms.

4.7 Diagrams

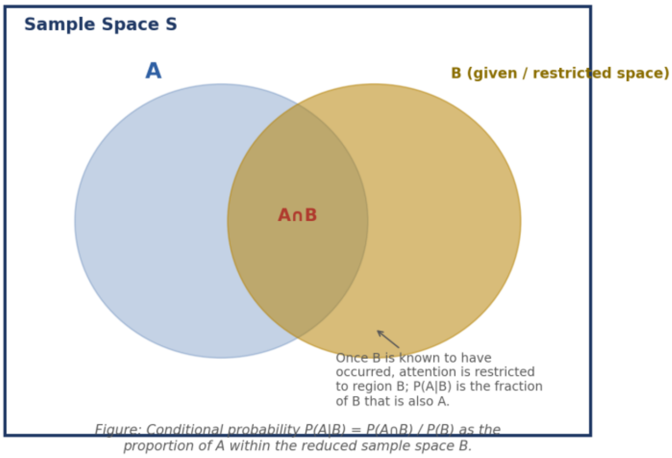

Figure 8: *Venn diagram for conditional probability. Once B is known to have occurred, the sample space is restricted to the region B, and P(A|B) is the proportion of B that is also covered by A.*

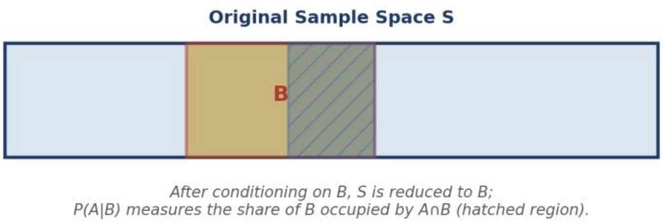

Figure 9: *Restriction of the sample space upon conditioning on event B. The hatched region A ∩ B, expressed as a fraction of the shaded region B, gives P(A|B).*

4.8 Solved Example — Question 1

(Original University-Level Question)

Question: A single fair die is rolled once. Let A be the event that the number obtained is greater than 3, and let B be the event that the number obtained is even. Find P(A|B).

Given Data

Sample space, S	{1, 2, 3, 4, 5, 6}
Event A: number > 3	{4, 5, 6}

Event B: number is even	{2, 4, 6}
Intersection, A ∩ B	{4, 6}

Solution

Step 1 — Compute P(B). Since S has 6 equally likely outcomes and B contains 3 of them:

$$P(B) = n(B)/n(S) = 3/6 = 1/2$$

Step 2 — Compute P(A ∩ B). The set A ∩ B = {4, 6} contains 2 outcomes, so:

$$P(A \cap B) = n(A \cap B)/n(S) = 2/6 = 1/3$$

Step 3 — Apply the conditional probability formula, substituting the two values computed above.

$$P(A|B) = P(A \cap B) \;/\; P(B) = (1/3) \;/\; (1/2) = (1/3) \times (2/1) = 2/3$$

This result can be verified directly by restricting attention to B = {2, 4, 6}: of these three equally likely outcomes, exactly two (namely 4 and 6) also satisfy A, giving P(A|B) = 2/3, which matches the formula-based calculation.

Final Answer: $P(A|B) = 2/3$

4.9 Solved Example — Question 2

(Adapted from GATE Probability Questions, GATE CS 2016)

Question: In a certain college, 60% of the students are enrolled in an Engineering programme, and 25% of the students are enrolled in both Engineering and a minor in Data Science. A student is selected at random and is found to be enrolled in Engineering. Find the probability that this student is also enrolled in the Data Science minor.

Given Data

P(Engineering) = P(E)	0.60
P(Engineering ∩ Data Science) = P(E ∩ D)	0.25

What is to be Found

The probability that a student is enrolled in the Data Science minor, given that the student is known to be enrolled in Engineering — that is, P(D|E).

Formula Used

$$P(D|E) = P(E \cap D) \;/\; P(E)$$

Solution

Step 1 — Identify the known values from the problem statement: $P(E) = 0.60$ and $P(E \cap D) = 0.25$.

Step 2 — Substitute these values directly into the conditional probability formula.

$$P(D|E) = 0.25 / 0.60$$

Step 3 — Simplify the resulting fraction. Dividing numerator and denominator by 0.05 gives 5/12, which as a decimal is approximately 0.4167.

$$P(D|E) = 25/60 = 5/12 \approx 0.4167$$

Reasoning: because the sample space has been restricted to students who are already known to be Engineering students (60% of the college), the conditional probability re-expresses the 25% overall figure as a fraction of that smaller, restricted group rather than of the whole college.

Final Answer: $P(D|E) = 5/12 \approx 0.417$ (41.7%)

Exam Tip: *Whenever a question states "given that…" or describes a subject already known to satisfy one condition, this is a direct verbal cue for conditional probability. Identify the conditioning event carefully — it always appears in the denominator.*

5. Total Probability Theorem

5.1 Introduction

Frequently, an event of interest can occur through several distinct, mutually exclusive routes or causes, and the probability of the event under each individual route is known, but the overall probability of the event across all routes is not directly given. The Total Probability Theorem provides a systematic method for combining these route-specific probabilities into a single overall probability. It is an indispensable stepping stone toward Bayes' Theorem, developed in Section 6.

5.2 Definition and Statement

Let $B_1, B_2, \ldots, B_n$ be a partition of the sample space S — that is, a collection of events that are pairwise mutually exclusive ($B_i \cap B_j = \emptyset$ for $i \neq j$) and collectively exhaustive ($B_1 \cup B_2 \cup \ldots \cup B_n = S$), each with $P(B_i) > 0$. For any event A defined on the same sample space, the Total Probability Theorem states that the overall probability of A can be obtained by summing the probability of A occurring jointly with each B_i.

5.3 Formula

$$P(A) = \Sigma_i\, P(B_i) \times P(A|B_i) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + \ldots + P(B_n)P(A|B_n)$$

5.4 Explanation

Because $B_1, B_2, \ldots, B_n$ partition the entire sample space, event A must occur together with exactly one of these B_i on any given trial — it is impossible for A to occur without simultaneously falling under one of the partitioning events. The theorem simply expresses this fact algebraically: the total probability of A is the sum, over every possible "cause" or "route" B_i, of the probability that both the route B_i and the event A occur together. Each such joint probability $P(B_i \cap A)$ is rewritten, using the Multiplication Rule from Section 3, as $P(B_i) \times P(A|B_i)$, which is why prior probabilities and conditional (likelihood) probabilities appear side by side in the formula.

5.5 Applications

- Computing the overall probability that a manufactured item is defective, when it may have been produced on any one of several machines, each with a different defect rate.
- Computing the overall probability of a positive medical test result across a population composed of both diseased and healthy sub-groups.
- Estimating the overall probability of network packet loss when traffic may be routed through any of several paths, each with a distinct loss probability.
- Serving as the normalising denominator, the "evidence" term, inside Bayes' Theorem itself.

5.6 Diagrams

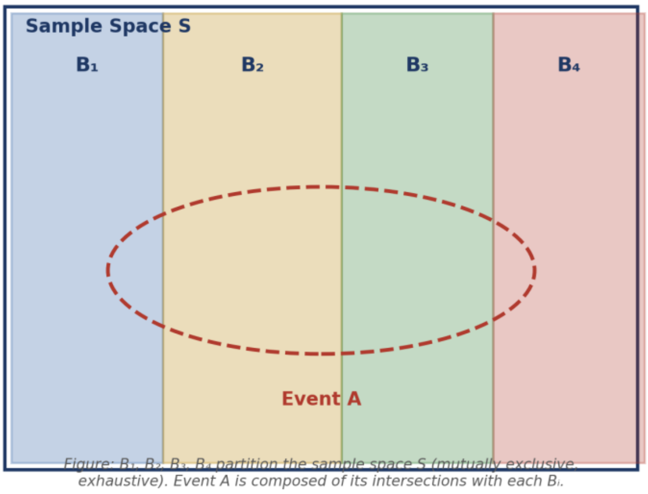

Figure 10: *Partition of the sample space S into mutually exclusive, exhaustive events B_1, B_2, B_3, B_4. Event A is distributed across the partition, and its total probability is the sum of its intersections with each B_i.*

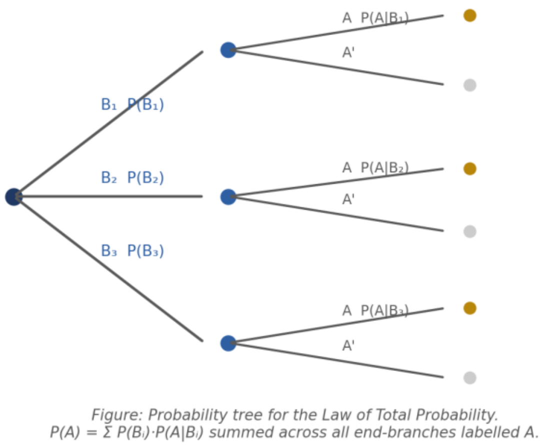

Figure 11: *Probability tree corresponding to the Total Probability Theorem. The overall probability of A is obtained by summing $P(B_i) \times P(A|B_i)$ over every branch that terminates in A.*

5.7 Solved Example — Question 1

(Original University-Level Question)

Question: A factory has three machines, M1, M2, and M3, which produce 25%, 35%, and 40% of the factory's total output respectively. From past records, it is known that 5% of the items produced by M1, 4% of the items produced by M2, and 2% of the items produced by M3 are defective. If an item is selected at random from the factory's total output, find the probability that it is defective.

Given Data

P(M1), P(M2), P(M3)	0.25 , 0.35 , 0.40
P(Defective | M1)	0.05
P(Defective | M2)	0.04
P(Defective | M3)	0.02

Formula Used

$$P(D) = P(M1)P(D|M1) + P(M2)P(D|M2) + P(M3)P(D|M3)$$

Solution

Step 1 — Compute the contribution of each machine separately by multiplying its production share by its own defect rate.

$$P(M1)\times P(D|M1) = 0.25 \times 0.05 = 0.0125$$

$$P(M2)\times P(D|M2) = 0.35 \times 0.04 = 0.0140$$

$$P(M3)\times P(D|M3) = 0.40 \times 0.02 = 0.0080$$

Step 2 — Sum the three contributions to obtain the overall probability that a randomly chosen item is defective, in accordance with the Total Probability Theorem.

$$P(D) = 0.0125 + 0.0140 + 0.0080 = 0.0345$$

Final Answer: P(Defective) = 0.0345 (3.45%)

5.8 Solved Example — Question 2

(Adapted from GATE Probability Questions, GATE CS 2019)

Question: A person owns three bags. Bag I contains 4 red and 6 black balls; Bag II contains 5 red and 5 black balls; Bag III contains 7 red and 3 black balls. A bag is chosen at random (each with equal probability of 1/3), and then a ball is drawn at random from the chosen bag. Find the probability that the ball drawn is red.

Given Data

P(Bag I) = P(Bag II) = P(Bag III)	1/3 each
P(Red | Bag I)	4/10 = 0.4
P(Red | Bag II)	5/10 = 0.5
P(Red | Bag III)	7/10 = 0.7

Formula Used

$$P(Red) = P(I)P(Red|I) + P(II)P(Red|II) + P(III)P(Red|III)$$

Solution

Step 1 — Compute each term of the sum, multiplying the (equal) prior probability of choosing each bag by that bag's conditional probability of drawing red.

$P(I) \times P(Red \mid I) = (1/3)(0.4) = 0.1333$
$P(II) \times P(Red \mid II) = (1/3)(0.5) = 0.1667$
$P(III) \times P(Red \mid III) = (1/3)(0.7) = 0.2333$

Step 2 — Add the three contributions to obtain the total, unconditional probability of drawing a red ball, regardless of which bag was chosen.

$$P(Red) = 0.1333 + 0.1667 + 0.2333 = 0.5333$$

Equivalently, working entirely in fractions: P(Red) = (1/3)(4/10 + 5/10 + 7/10) = (1/3)(16/10) = 16/30 = 8/15 ≈ 0.5333, which confirms the decimal calculation above.

Final Answer: $\mathbf{P(Red) = 8/15 \approx 0.5333}$

Exam Tip: *When every "cause" (machine, bag, route) is equally likely, factor the common prior probability out of the summation — this often turns three multiplications into one, reducing the chance of arithmetic slips under exam time pressure.*

6. Bayes' Theorem

6.1 Introduction

The Total Probability Theorem of Section 5 answers a "forward" question: given the probability of each cause and the probability of an effect under each cause, what is the overall probability of the effect? Bayes' Theorem, named after the eighteenth-century clergyman and mathematician Thomas Bayes, answers the corresponding "backward" question: given that the effect has already been observed, what is the probability that it arose from a particular cause? This inversion — from cause-to-effect reasoning to effect-to-cause reasoning — is one of the most powerful ideas in the whole of probability theory, and it forms the mathematical backbone of the Bayesian classification techniques discussed in Section 7.

6.2 Definition

Bayes' Theorem provides a formula for computing the conditional probability of an event B_i, drawn from a partition $B_1, B_2, \ldots, B_n$ of the sample space, given that another event A has already been observed to occur.

6.3 Formula

$$P(B_i|A) = [\, P(B_i) \times P(A|B_i) \,] \;/\; \Sigma_j [\, P(B_j) \times P(A|B_j) \,]$$

Recognising that the denominator is exactly the Total Probability Theorem expression for P(A) derived in Section 5, the formula is often written more compactly as:

$$P(B_i|A) = [\, P(B_i) \times P(A|B_i) \,] \;/\; P(A)$$

6.4 Explanation of Terms

Every application of Bayes' Theorem involves four distinct quantities, each of which must be correctly identified before the formula can be applied.

6.4.1 Prior Probability — $P(B_i)$

The prior probability is the probability assigned to the cause or hypothesis B_i before any new evidence is taken into account. It reflects existing knowledge, historical data, or a baseline assumption.

6.4.2 Likelihood — $P(A|B_i)$

The likelihood is the conditional probability of observing the evidence A, assuming that the particular cause B_i is true. It measures how well each candidate cause "explains" the observed evidence.

6.4.3 Evidence — P(A)

The evidence, also called the marginal probability of A, is the overall probability of observing A, averaged across every possible cause, exactly as computed by the Total Probability Theorem. It acts purely as a normalising constant, ensuring that the resulting posterior probabilities across all B_i sum to 1.

6.4.4 Posterior Probability — $P(B_i|A)$

The posterior probability is the updated, revised probability of the cause B_i after the evidence A has been taken into account. It is the quantity that Bayes' Theorem ultimately computes, and it combines the prior belief with the strength of the observed evidence.

6.5 Applications

- Medical diagnosis: revising the probability that a patient has a disease, given the result of a diagnostic test.
- Spam filtering: revising the probability that an email is spam, given the specific words it contains — the direct precursor to the Naïve Bayes Classifier of Section 7.
- Legal and forensic reasoning: updating the probability of a suspect's guilt, given newly presented evidence.
- Quality control: identifying the most probable source machine for a defective item, given that a defect has been observed.
- Search-and-rescue and target-tracking systems, which continuously update location probabilities as new sensor readings arrive.

6.6 Diagrams

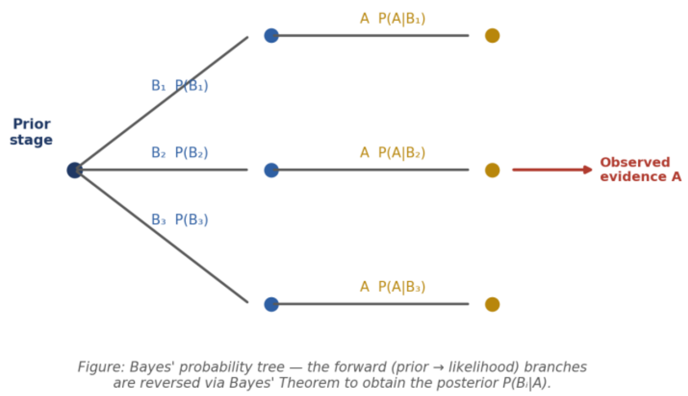

Figure 12: *Bayes' probability tree. The forward branches represent the prior $P(B_i)$ and likelihood $P(A|B_i)$; Bayes' Theorem reverses these branches to compute the posterior $P(B_i|A)$ once the evidence A has been observed.*

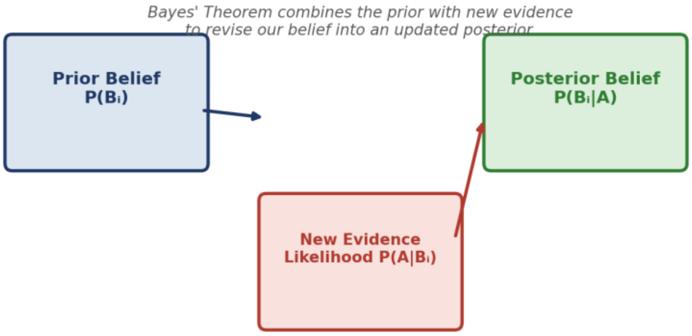

Figure 13: *Bayesian updating diagram, showing how a prior belief is combined with new evidence (the likelihood) to produce a revised posterior belief.*

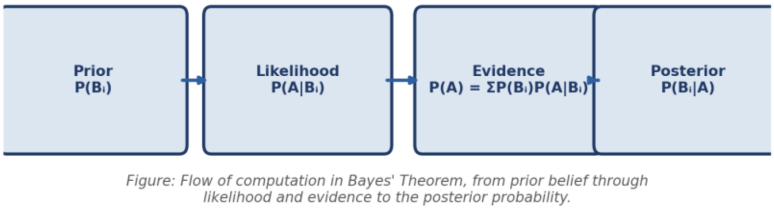

Figure 14: *Flow diagram summarising the computation of Bayes' Theorem, from prior probability through likelihood and evidence to the final posterior probability.*

6.7 Solved Example — Question 1

(Original University-Level Question)

Question: Two urns are used in an experiment. Urn A contains 3 red balls and 7 white balls; Urn B contains 6 red balls and 4 white balls. An urn is chosen at random with equal probability, and a ball is drawn from it, which turns out to be red. Find the probability that the chosen urn was Urn A.

Given Data

P(Urn A) = P(Urn B)	1/2 each
P(Red | Urn A)	3/10 = 0.3
P(Red | Urn B)	6/10 = 0.6

What is to be Found

The posterior probability P(Urn A | Red) — that is, given that a red ball was drawn, the probability that it came from Urn A.

Formula Used

$$P(A|Red) = [P(A)\times P(Red|A)] \;/\; [P(A)\times P(Red|A) + P(B)\times P(Red|B)]$$

Solution

Step 1 — Compute the numerator, the joint probability that Urn A is chosen and a red ball is drawn from it.

$$P(A) \times P(\text{Red}|A) = 0.5 \times 0.3 = 0.15$$

Step 2 — Compute the corresponding joint probability for Urn B, which is required for the denominator (the total evidence).

$$P(B) \times P(\text{Red}|B) = 0.5 \times 0.6 = 0.30$$

Step 3 — Sum the two joint probabilities to obtain P(Red), the overall (unconditional) probability of drawing a red ball, by the Total Probability Theorem.

$$P(\text{Red}) = 0.15 + 0.30 = 0.45$$

Step 4 — Divide the numerator obtained in Step 1 by the total evidence obtained in Step 3, in accordance with Bayes' Theorem.

$$P(A|\text{Red}) = 0.15 \,/\, 0.45 = 1/3 \approx 0.3333$$

Interpretation: although Urn A was chosen with the same prior probability as Urn B, the fact that a red ball was drawn provides evidence favouring Urn B, since Urn B contains a higher proportion of red balls. Consequently, the posterior probability of Urn A (1/3) is lower than its prior probability (1/2).

Final Answer: P(Urn A | Red) = 1/3 ≈ 0.333

Exam Tip: *After computing a posterior probability, always compare it with the corresponding prior. If the evidence favours a different hypothesis, the posterior should be lower than the prior — this is a quick sanity check against arithmetic errors.*

6.8 Solved Example — Question 2

(Classic Probability Question — Truth-Teller and Die Problem)

Question: A man is known to speak the truth three out of four times. He throws a die and reports that it is a six. Find the probability that the die actually showed a six.

Given Data

Let T denote the event that the man speaks the truth, and R denote the event that he reports the die showed a six.

P(Man speaks truth), P(T)	3/4
P(Man lies), P(T′)	1 − 3/4 = 1/4
P(Die actually shows six), P(Six)	1/6 (one favourable face out of six)

P(Die does not show six), P(Six′)	5/6

What is to be Found

The posterior probability that the die actually showed a six, given that the man reported a six — that is, P(Six | R).

Formula Used

Two hypotheses partition the sample space: either the die genuinely showed a six, or it did not. The report R ("he says it is a six") is truthful in the first case with probability P(T), and is a lie that happens to claim "six" in the second case with probability P(T′) × (1/5), since if the die showed some other number and the man lies, he could name any one of the five remaining faces, only one of which is "six".

$$P(Six|R) = [P(Six)\times P(T)] \;/\; [P(Six)\times P(T) + P(Six')\times P(T')\times(1/5)]$$

Solution

Step 1 — Compute the numerator: the probability that the die shows six AND the man truthfully reports six.

$$P(Six) \times P(T) = (1/6) \times (3/4) = 3/24 = 1/8$$

Step 2 — Compute the probability that the die does NOT show six, the man lies, and his lie happens to be exactly "six" (one out of the five untrue faces he could have named).

$$P(Six') \times P(T') \times (1/5) = (5/6) \times (1/4) \times (1/5) = 5/120 = 1/24$$

Step 3 — Add the results of Steps 1 and 2 to obtain the total probability of the evidence R (the man reporting "six"), regardless of what the die actually showed.

$$P(R) = 1/8 + 1/24 = 3/24 + 1/24 = 4/24 = 1/6$$

Step 4 — Apply Bayes' Theorem, dividing the numerator from Step 1 by the total evidence from Step 3.

$$P(Six|R) = (1/8) \;/\; (1/6) = (1/8) \times 6 = 6/8 = 3/4$$

Interpretation: despite the die having only a 1-in-6 chance of showing six in the first place, the man's high truthfulness (3/4) raises the posterior probability considerably, to 3/4, once his report is taken into account.

Final Answer: P(Die actually showed six | Report of six) = 3/4 = 0.75

> ***Exam Tip:*** *In truth-teller problems, always account for the fact that a liar naming a specific wrong face is only one of several possible lies — this is the origin of the (1/5) factor, and omitting it is the single most common mistake in this classic question.*

6.9 Solved Example — Question 3

(Adapted from GATE Probability Questions, GATE CS 2021 — Difficult Level)

Question: In a certain region, it is known that 1% of the population suffers from a rare genetic disorder. A diagnostic test for this disorder is 98% accurate when the disorder is actually present (true positive rate), and produces a false positive in 3% of cases where the disorder is absent. If a person selected at random from the population tests positive, find the probability that the person actually has the disorder.

Given Data

P(Disorder present), P(D)	0.01
P(Disorder absent), P(D′)	0.99
P(Positive | D) (true positive rate)	0.98
P(Positive | D′) (false positive rate)	0.03

What is to be Found

The posterior probability that a person actually has the disorder, given a positive test result — that is, P(D | Positive).

Formula Used

$$P(D|Pos) = [P(D)\times P(Pos|D)] \,/\, [P(D)\times P(Pos|D) + P(D')\times P(Pos|D')]$$

Solution

Step 1 — Compute the numerator: the joint probability that a person has the disorder AND correctly tests positive.

$$P(D) \times P(Pos|D) = 0.01 \times 0.98 = 0.0098$$

Step 2 — Compute the joint probability that a person does NOT have the disorder but nonetheless tests positive (a false positive).

$$P(D') \times P(Pos|D') = 0.99 \times 0.03 = 0.0297$$

Step 3 — Add the two joint probabilities to obtain P(Positive), the total, unconditional probability that a randomly selected person tests positive, whether or not they actually have the disorder.

$$\mathrm{P(Positive)} = 0.0098 + 0.0297 = 0.0395$$

Step 4 — Divide the numerator (Step 1) by the total evidence (Step 3) to obtain the posterior probability.

$$\mathrm{P(D|Positive)} = 0.0098 \,/\, 0.0395 \approx 0.2481$$

Reasoning: although the diagnostic test is highly accurate (98% true positive rate, only 3% false positive rate), the disorder itself is very rare in the population (only 1%). Consequently, the small number of genuine positive cases is heavily outweighed, in absolute terms, by the comparatively large number of false positives arising from the 99% of the population who do not have the disorder. This counter-intuitive outcome is known as the base-rate effect, and it is one of the most important practical lessons of Bayesian reasoning in medical and diagnostic contexts.

Final Answer: P(Disorder | Positive test) ≈ 0.248 (about 24.8%)

Exam Tip: *Whenever a GATE question combines a low prior (rare disease, rare fault) with an imperfect test, expect the posterior probability to be surprisingly low — this is the base-rate fallacy trap, and Bayes' Theorem is precisely the tool that avoids it.*

7. Bayesian Classification

7.1 Introduction

Bayesian classification is the application of Bayes' Theorem, developed in Section 6, to the problem of supervised machine learning — specifically, to the task of assigning a class label to a new, previously unseen data instance on the basis of its observed features. Rather than deterministically computing a class, a Bayesian classifier computes the posterior probability of each candidate class given the observed evidence, and then selects the class with the highest posterior probability.

7.2 Definition

A Bayesian classifier is a probabilistic model that predicts the class of an instance by treating the class label as a hypothesis and the observed feature values as evidence, applying Bayes' Theorem to compute the posterior probability of each class, and outputting the class that maximises this posterior probability — a decision rule known as Maximum A Posteriori (MAP) estimation.

7.3 Historical Note

Although Bayes' Theorem itself dates to the 1760s, its application to automated pattern classification developed alongside the broader fields of statistical pattern recognition and machine learning during the twentieth century. The Naïve Bayes classifier, in particular, gained widespread popularity from the 1990s onward owing to its remarkable effectiveness in text classification tasks, most notably email spam filtering, despite its simplifying independence assumption.

7.4 Bayes' Theorem in Machine Learning

In the classification setting, Bayes' Theorem is rewritten with C representing a class label and x representing a feature vector $(x_1, x_2, \ldots, x_n)$ describing the instance to be classified.

$$P(C|x) = [P(C) \times P(x|C)] \,/\, P(x)$$

Here, $P(C)$ is the prior probability of the class (its overall frequency in the training data), $P(x|C)$ is the likelihood of observing the feature vector given the class, $P(x)$ is the evidence — a normalising constant common to every class — and $P(C|x)$ is the posterior probability that determines the final prediction.

7.5 Naïve Bayes Classifier

The direct computation of $P(x|C)$ is generally intractable, because it requires estimating the joint distribution of all n features conditioned on the class, which grows exponentially with n. The Naïve Bayes classifier resolves this difficulty by making a simplifying assumption.

7.5.1 Assumptions of Naïve Bayes

- Conditional independence: given the class label, every feature x_i is assumed to be statistically independent of every other feature x_j.
- Equal importance: in its basic form, every feature is treated as contributing independently to the classification decision, without inherent weighting.

Under this assumption, the joint likelihood P(x|C) simplifies to a simple product of individual per-feature likelihoods, which can be estimated efficiently from training data.

$$P(x|C) = P(x_1|C) \times P(x_2|C) \times \ldots \times P(x_n|C) = \Pi_i \, P(x_i|C)$$

7.5.2 Types of Naïve Bayes

- Gaussian Naïve Bayes: assumes that continuous-valued features, within each class, follow a normal (Gaussian) distribution; used for real-valued measurements such as sensor readings.
- Multinomial Naïve Bayes: models feature counts (such as word frequencies in a document) using a multinomial distribution; the standard choice for text classification.
- Bernoulli Naïve Bayes: models binary features (such as the presence or absence of a specific word) using a Bernoulli distribution; suited to short-text and presence/absence problems.

7.6 Working Principle

7.6.1 Mathematical Formula

Combining the independence assumption with Bayes' Theorem, and noting that P(x) is identical across all candidate classes and therefore does not affect which class has the highest posterior, the classifier's decision rule reduces to selecting the class $\hat{C}$ that maximises the numerator alone.

$$\hat{C} = \text{argmax over C of } [\, P(C) \times \Pi_i \, P(x_i|C) \,]$$

7.6.2 Training Process

During training, the classifier estimates the prior probability P(C) for each class as the relative frequency of that class in the training set, and estimates each conditional feature probability $P(x_i|C)$ from the training instances belonging to that class — for example, by computing the sample mean and variance for Gaussian Naïve Bayes, or by computing relative word frequencies for Multinomial Naïve Bayes.

7.6.3 Prediction Process

Given a new, unlabelled instance, the classifier computes the product $P(C) \times \Pi_i \, P(x_i|C)$ for every candidate class, and assigns the instance to the class for which this product is largest. In practice, because multiplying many small probabilities can cause numerical underflow, implementations typically work with the sum of log-probabilities rather than the raw product.

7.7 Diagrams

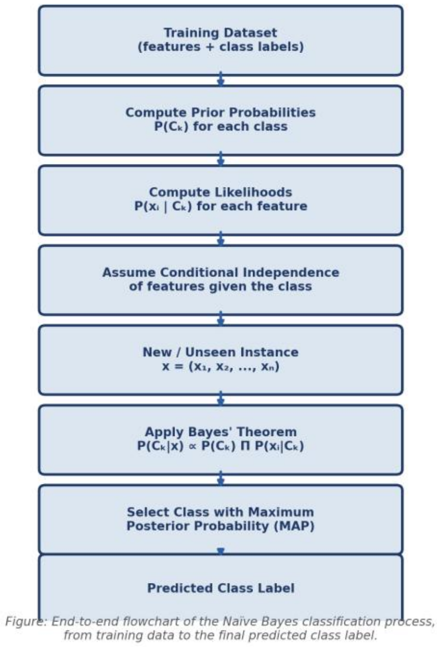

Figure 15: *End-to-end flowchart of the Naïve Bayes classification process, from the training dataset through prior and likelihood estimation to the final predicted class label.*

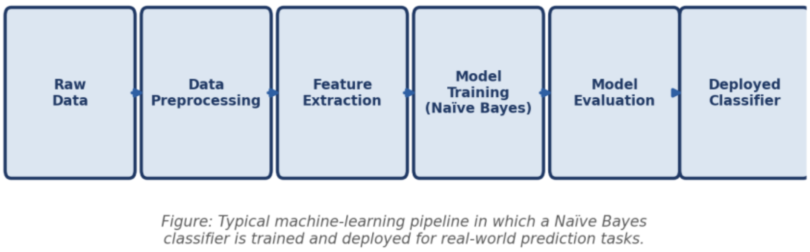

Figure 16: *Typical machine-learning pipeline showing where a Naïve Bayes classifier fits within the broader workflow, from raw data collection to a deployed classifier.*

7.8 Advantages

- Computationally efficient to train and to use for prediction, scaling linearly with the number of features and training instances.
- Performs well even with comparatively small training datasets.
- Handles high-dimensional data, such as text represented by thousands of word-features, gracefully.
- Naturally supports multi-class classification without modification.

- Remains surprisingly competitive in accuracy even when the independence assumption is violated in practice.

7.9 Disadvantages

- The conditional independence assumption is rarely exactly true in real data, which can degrade probability estimates even when the final classification decision remains correct.
- A feature value that never appears with a given class in the training data assigns that class a likelihood of exactly zero (the "zero-frequency problem"), typically addressed using Laplace (additive) smoothing.
- Continuous features must be modelled using an assumed distribution (commonly Gaussian), which may not accurately reflect the true underlying distribution of the data.

7.10 Real-world Applications

- Email Spam Detection: classifying incoming emails as "spam" or "not spam" based on the words and patterns they contain.
- Sentiment Analysis: classifying text such as product reviews or social media posts as expressing positive, negative, or neutral sentiment.
- Medical Diagnosis: estimating the probability of a disease given a set of observed symptoms or test results.
- Document Classification: automatically sorting news articles, support tickets, or research papers into predefined topic categories.
- Fraud Detection: flagging financial transactions as potentially fraudulent based on patterns in transaction features.

7.11 Conclusion

Bayesian classification demonstrates how the abstract mathematics of Bayes' Theorem translates directly into a practical, widely deployed machine-learning technique. By treating classification as a problem of computing posterior probabilities over candidate classes, and by adopting the simplifying but remarkably effective assumption of conditional feature independence, the Naïve Bayes classifier achieves an appealing balance of simplicity, computational efficiency, and predictive performance. It thereby serves as a natural bridge between the theoretical foundations of probability covered in the earlier sections of this assignment and the applied domain of data analytics and machine learning.

> ***Exam Tip:*** *When a question asks you to justify why Naïve Bayes is called "naïve", always point specifically to the conditional independence assumption among features — this is the single most commonly tested conceptual point in university and GATE-style examinations on this topic.*

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